1. Strong number.

A strong number is a number in which the sum of the factorials of its digits equals the number itself.

Defination :-

A number N is a strong number if :

Sum of factorial of digits of N = N.

Examples :-

Number	Calculation	Strong?
145	$1!+4!+5! = 1+24+120 = 145$	Yes
2	$2! = 2$.	Yes
405	$4!+0!+5!+8!+5! = 40585$	Yes.

NOT strong example

123 →

1!+2!+3! = 1+2+6 = 9 ≠ 123.

python program to check strong number.

```
Num = int[input("enter a number")]
    s = 0
  Temp = num
  While temp >0;
 digit = temp % 10.
   fact = 1.
for i in range (1, digit +1):
   fact * = i
   s+ = fact
   Temp // = 10

If s= num;
printf("num, "strong is a number")
else ;
printf(num, " is not a strong number").
 return 0;
}
```

Enter a number : 145

145 is a strong number.

2. Perfect number :-

A perfect number is a number that is equal to the sum of it's proper divisions.

28 → Divisors : 1, 2, 7, 4, 14 → sum = 28.

```
# include < stdio.h>.
 int main ( )
{
int num , sum = 0 ;
printf ("Enter a number ") ;
scanf("%d", & num) ;
for (int i = 1; i < = num /2 ; i++)
{
 if (num % i = 0 )
{
 sum + = i ;
}
}
 if (sum = = num)
printf("%d is a Perfect number \n", num) ;
else
 printf("%d is NOT a perfect number \n", num) ;
 return 0 ;
}.
```

Output :-

Enter a number r = 28.

28 is a perfect number.